

L#30: Phasor and circuit elements

Impedance and admittance

Impedance (Z): the ratio between the phasor voltage and the phasor current.

$$Z = \frac{\bar{U}}{\bar{I}} = \underset{\uparrow}{R} + j \underset{\uparrow \text{ reactance.}}{X}$$

R: resistance. $R \geq 0$.

X: reactance. $X > 0$, inductance
 $X < 0$, capacitance

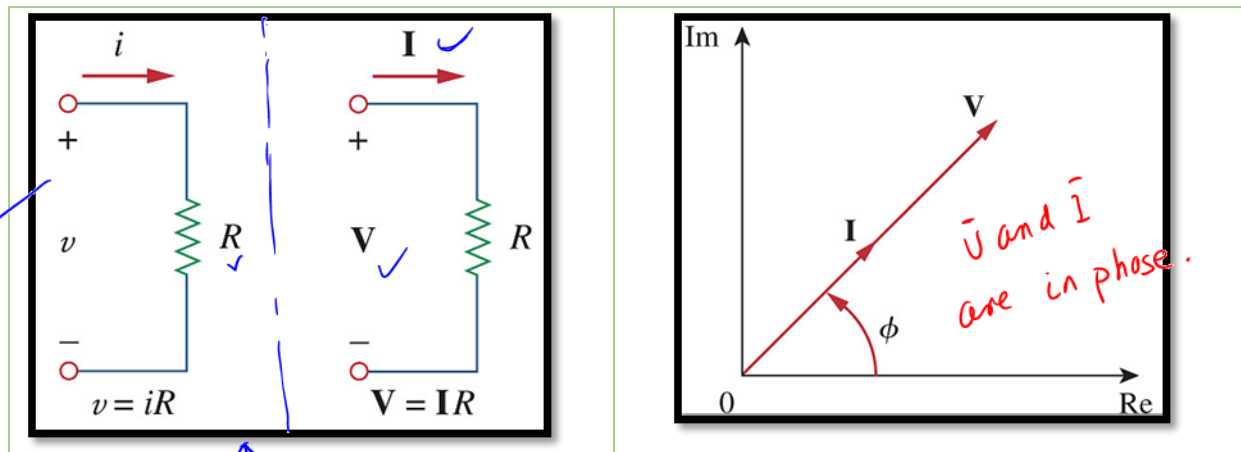
Admittance (Y):

as the inverse of impedance.

$$Y = \frac{1}{Z}$$

$$R = \frac{V}{I}, G = \frac{1}{R}$$
$$Z = \frac{\bar{V}}{\bar{I}}, Y = \frac{1}{Z}$$

Phasor and circuit elements - resistors



$$i(t) = I_m \cos(\omega t + \phi)$$

Given value of $i(t)$ and R , find Z_R .

① phasor voltage and phasor current.

$$\bar{I} = I_m \angle \phi.$$

$$v(t) = i(t) \cdot R = I_m R \cos(\omega t + \phi).$$

$$\bar{V} = I_m R \angle \phi.$$

$$\textcircled{2} \quad Z_R = \frac{\bar{V}}{\bar{I}} = \frac{I_m R \angle \phi}{I_m \angle \phi} = R \angle 0^\circ. \quad (1)$$

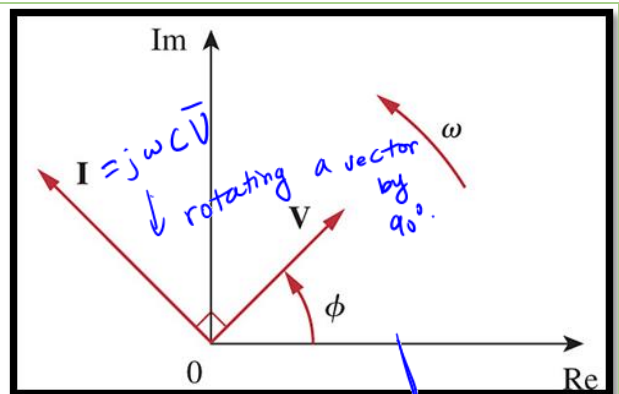
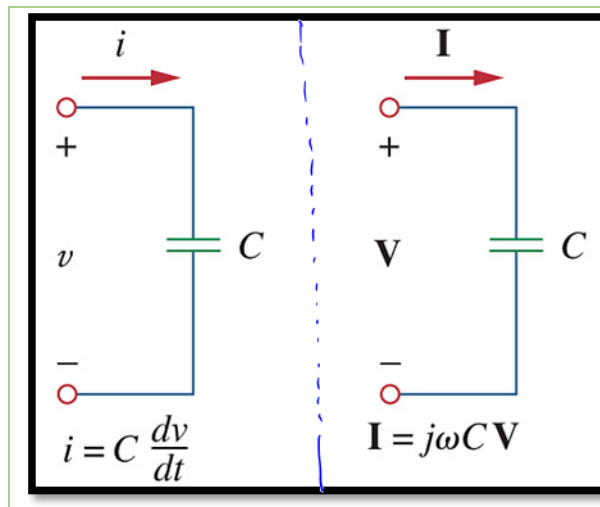
$$Z_R = R, \quad (1) \Rightarrow \bar{V} = \bar{I} R \angle 0^\circ \quad |\angle 0^\circ = 1$$

$$\bar{V} = \bar{I} R.$$

no phase shift
between $\bar{V}$ and $\bar{I}$.

- when convert signals from the time domain to phasor domain.
 - sign convention still applies.
 - Ohm's law still applies, $\bar{V} = \bar{I} \cdot Z_R$.

Phasor and circuit elements - capacitors



I leads V by 90° for a capacitor.

$$v(t) = V_m \cos(\omega t + \phi)$$

$$i(t) = C \frac{dv}{dt} = -C V_m \sin(\omega t + \phi) \cdot \omega$$

$$= -\omega C V_m \sin(\omega t + \phi)$$

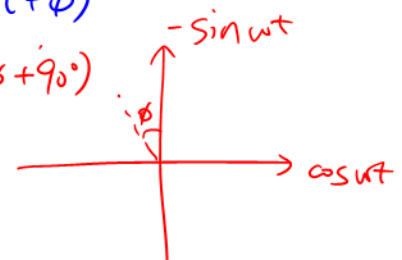
$$= \omega C V_m \cos(\omega t + \phi + 90^\circ)$$

$$\bar{V} = V_m \angle \phi, \quad \bar{I} = \omega C V_m \angle (\phi + 90^\circ)$$

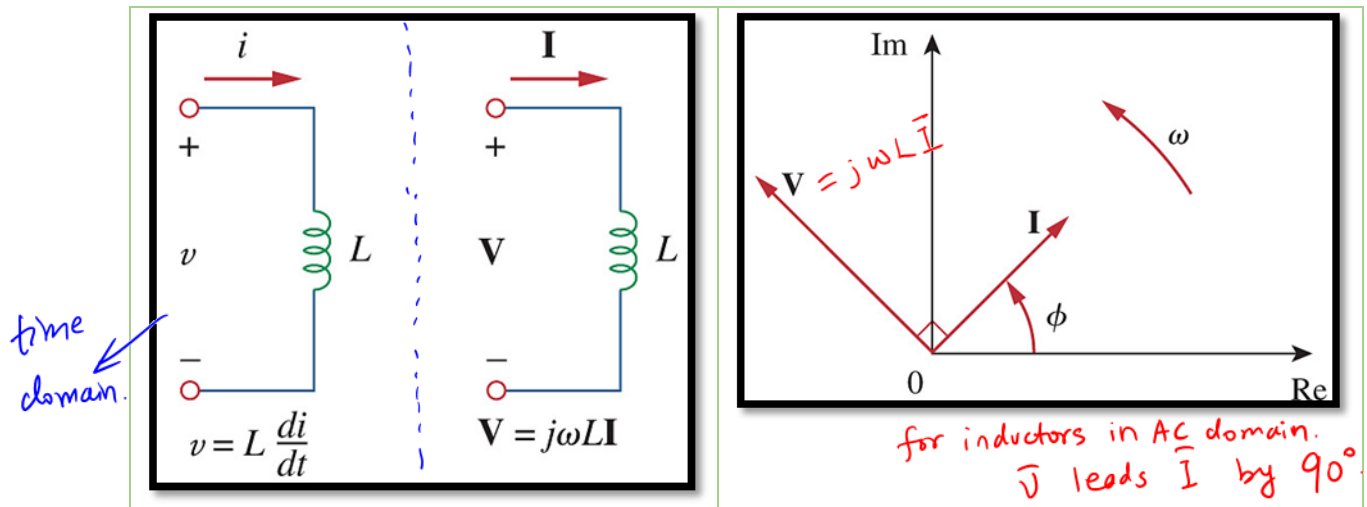
$$= j\omega C V_m \angle \phi$$

$$Z_C = \frac{\bar{V}}{\bar{I}} = \frac{V_m \angle \phi}{j\omega C V_m \angle \phi} = \boxed{\frac{1}{j\omega C}} = \frac{-j^2}{j\omega C} = \boxed{-j \frac{1}{\omega C}}$$

$$\bar{I} = j\omega C \underline{V_m \angle \phi} = j\omega C \bar{V}$$



Phasor and circuit elements - inductors



apply $i(t) = I_m \cos(\omega t + \phi)$ to L .

$$\begin{aligned} v(t) &= L \frac{di}{dt} = L I_m \cdot -\sin(\omega t + \phi) \cdot \omega \\ &= -\omega L I_m \sin(\omega t + \phi) \\ &= \omega L I_m \cos(\omega t + \phi + 90^\circ) \end{aligned}$$

$$\bar{I} = I_m \angle \phi, \quad \bar{V} = \omega L I_m \angle (\phi + 90^\circ)$$

$$= j\omega L I_m \angle \phi$$

$$\underline{Z_L} = \frac{\bar{V}}{\bar{I}} = \frac{j\omega L I_m \angle \phi}{I_m \angle \phi} = j\omega L$$

$$\bar{V} = j\omega L \bar{I}$$

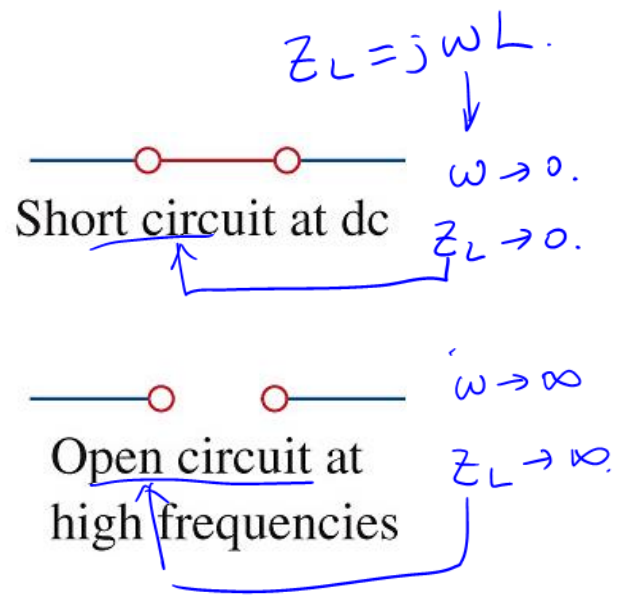
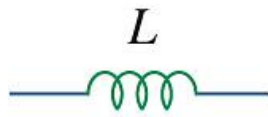
AC in phasor-domain frequency

$$\boxed{C \quad I \quad V \quad I \quad L}$$

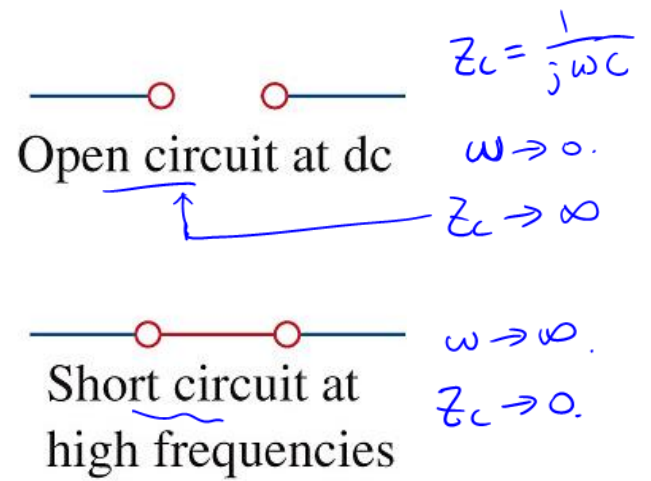
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Summary of voltage-current relationship		
Element	Time domain	Frequency domain
R	$v = i R$	$\bar{V} = \bar{I} R.$
L	$v_L = L \frac{di_L}{dt}$	$\bar{V}_L = j\omega L \bar{I}_L$
C	$i_C = C \frac{dv_C}{dt}$	$\bar{I}_C = j\omega C \bar{V}_C$

Impedances and admittances of passive elements		
Element	Impedance	Admittance
<u>R</u>	$Z_R = R \text{ } [\Omega]$	$Y_R = \frac{1}{R} \text{ } [S].$
L	$Z_L = j\omega L \text{ } [\Omega]$	$Y_L = \frac{1}{Z_L} = \frac{1}{j\omega L} \text{ } [S].$
C	$Z_C = \frac{1}{j\omega C} \text{ } [\Omega]$	$Y_C = \frac{1}{Z_C} = j\omega C \text{ } [S].$



(a)

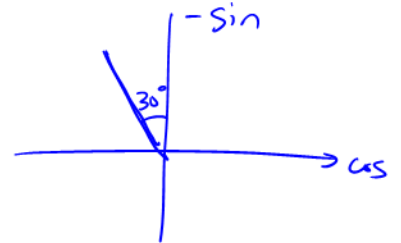


(b)

Example 5: phasor and impedance

If voltage $v = 10 \cos(\underbrace{100t}_{\omega} + 30^\circ)$ V is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the capacitor.

$$\begin{aligned} 1. \quad i_c &= C \frac{dv}{dt} = 50 \times 10^{-6} \times 10 \times -\sin(100t + 30^\circ) \times 100 \\ &= -5 \times 10^{-2} \sin(100t + 30^\circ) \\ &= \underline{0.05 \cos(100t + 120^\circ) \text{ A.}} \end{aligned}$$

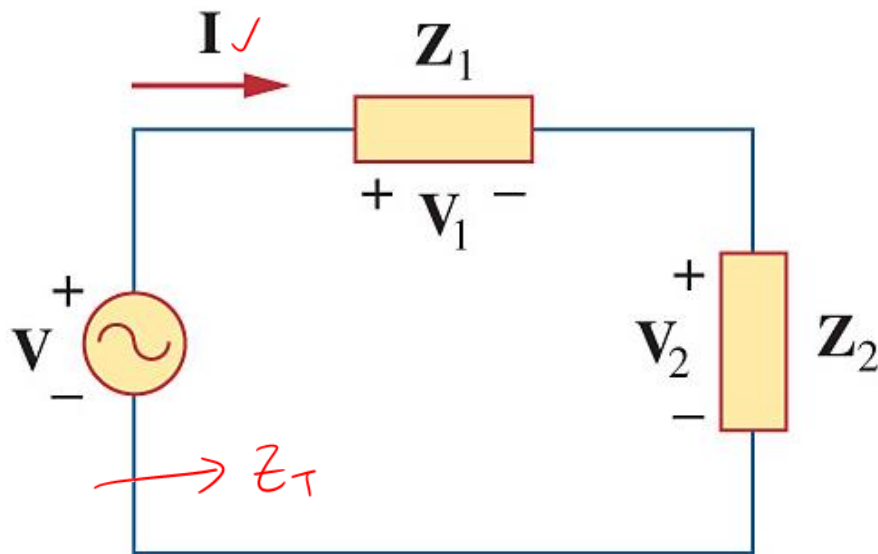


$$2. \quad \bar{V} = 10 \angle 30^\circ \text{ V.}$$

$$\left\{ \begin{aligned} Z_c &= \frac{1}{j\omega C} = \frac{1}{j \times 100 \times 50 \times 10^{-6}} = \frac{10^3}{j 5} = -j 200 \Omega. \\ \bar{I} &= \frac{\bar{V}}{Z_c} = \frac{10 \angle 30^\circ}{-j 200} = \frac{10 \angle 30^\circ}{200 \angle -90^\circ} = 0.05 \angle 120^\circ \text{ A.} \end{aligned} \right.$$

$$\underline{i(t) = 0.05 \cos(100t + 120^\circ) \text{ A.}}$$

AC circuit laws and theories

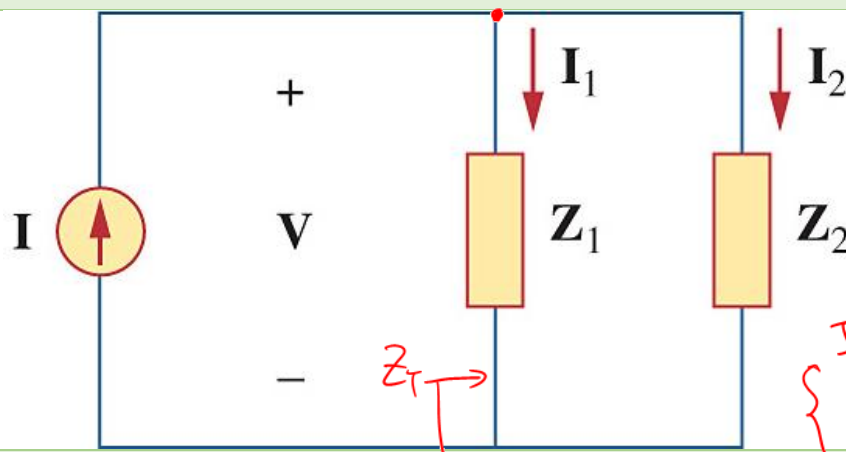


(KVL, Voltage division, Z)

$$\text{KVL: } -V + V_1 + V_2 = 0$$

$$Z_T = Z_1 + Z_2 \quad \bar{V}_1 = \bar{I} Z_1, \quad \bar{V}_2 = \bar{I} Z_2$$

$$V_1 = \bar{V} \cdot \frac{Z_1}{Z_1 + Z_2}, \quad V_2 = \bar{V} \cdot \frac{Z_2}{Z_1 + Z_2}$$



(KCL, current division, Z)

$$I = I_1 + I_2$$

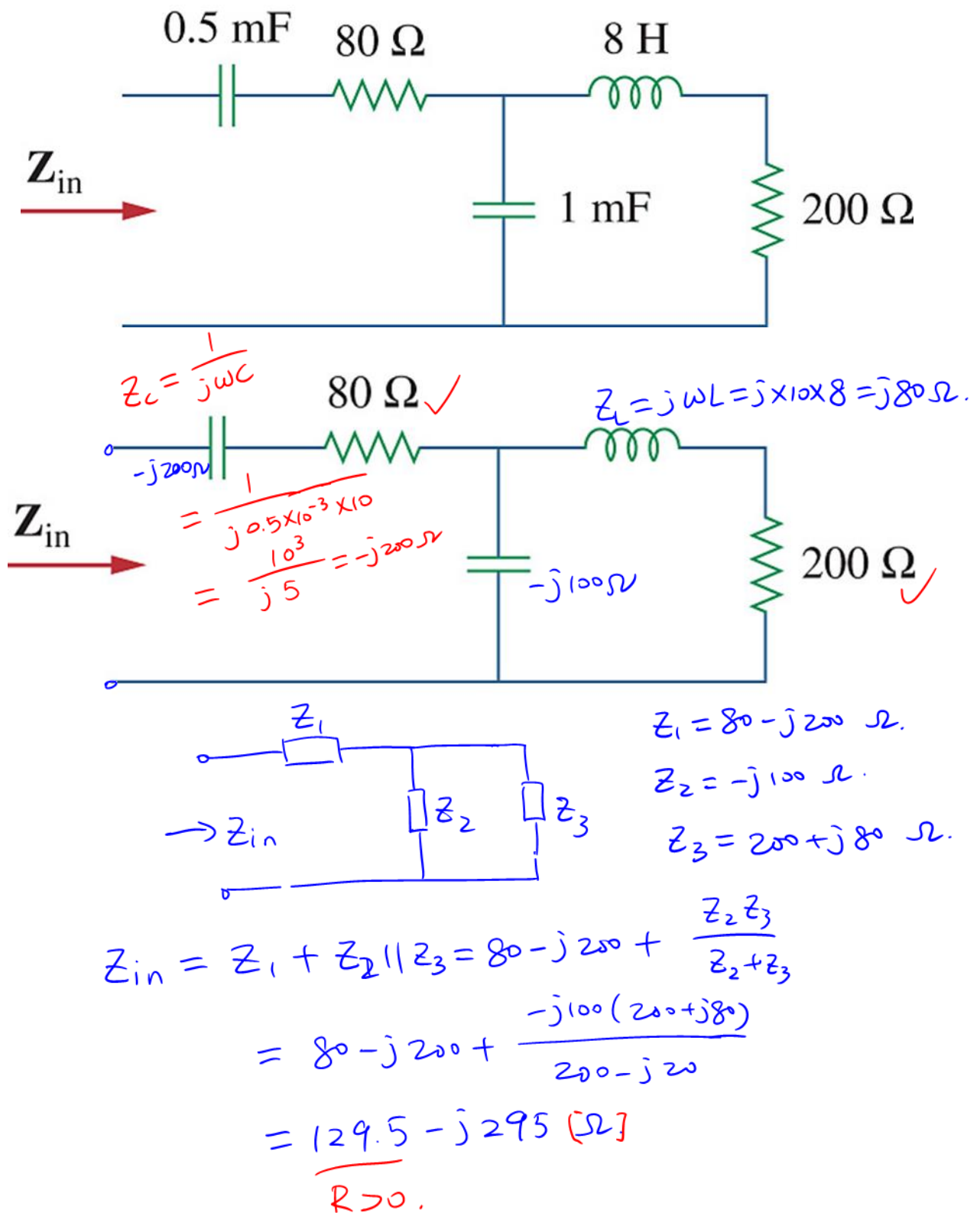
$$I_1 = \frac{\bar{V}}{Z_1}, \quad I_2 = \frac{\bar{V}}{Z_2}$$

$$I = \frac{\bar{V}}{Z_T} = \frac{\bar{V}}{Z_1} + \frac{\bar{V}}{Z_2} \Rightarrow \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$\Rightarrow Z_T = \frac{Z_1 Z_2}{Z_1 + Z_2}$$

Example 6: impedance

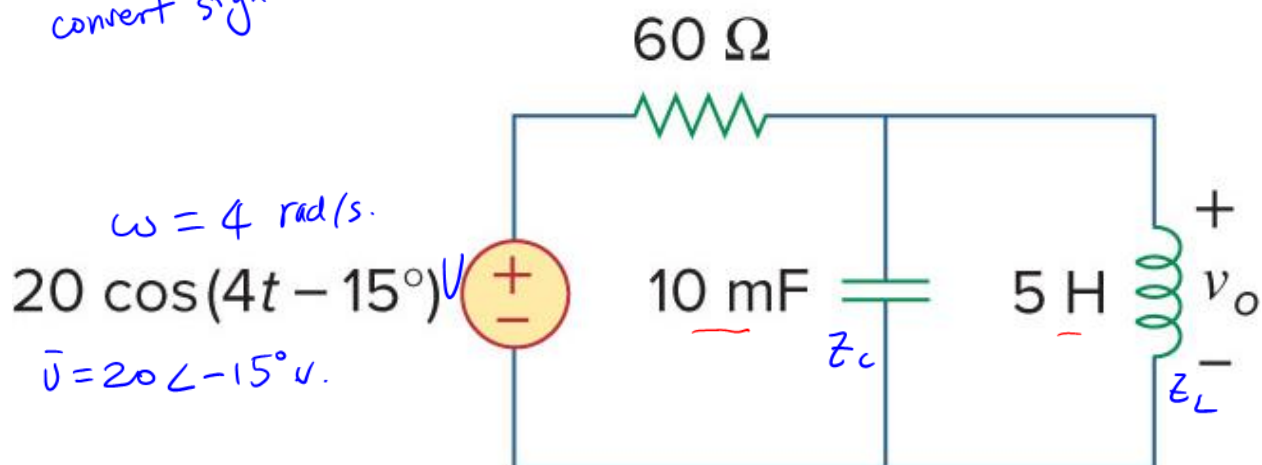
Determine the input impedance of the circuit at $\omega = 10$ rad/s.



Example 7: current and voltage divider

Find v_o .

convert signals into phasor domain.



$$Z_c = \frac{1}{j\omega C} = -j \frac{1}{\omega C} = -j \frac{1}{4 \times 10 \times 10^{-3}} = -j 25 \Omega.$$

$$Z_L = j\omega L = j \times 4 \times 5 = j 20 \Omega.$$

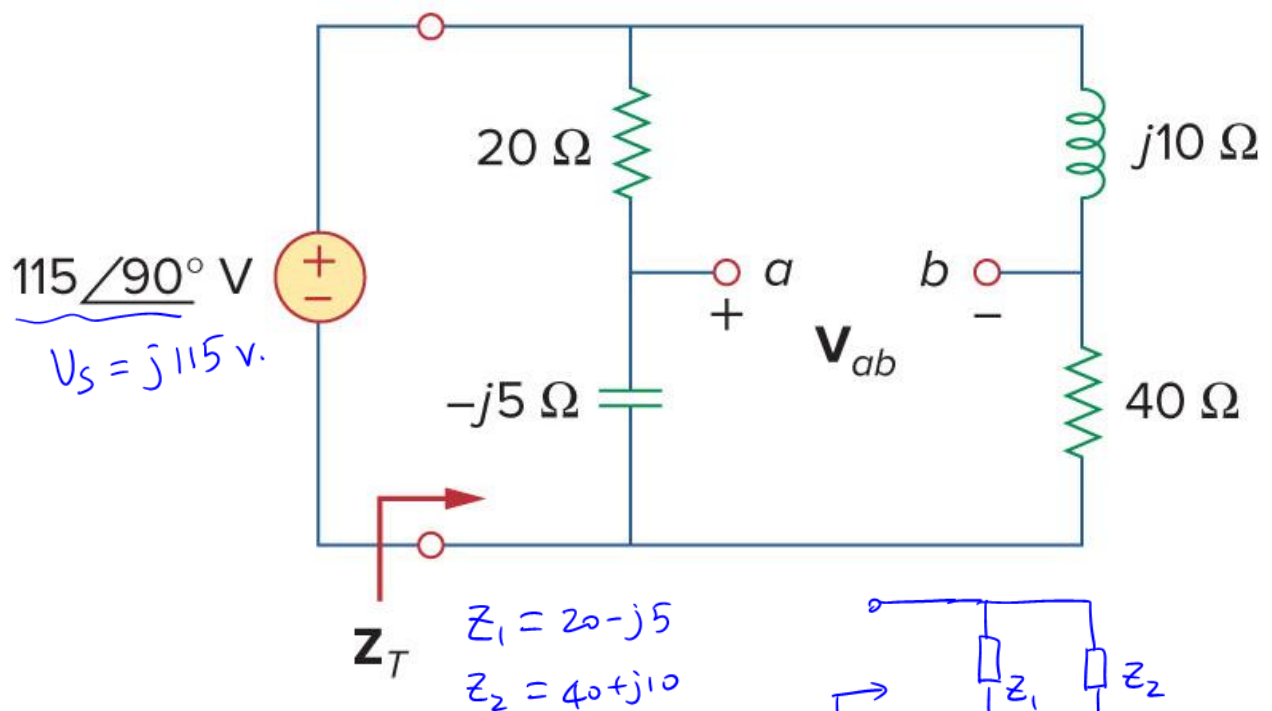
$$Z_1 = 60 \Omega, \quad Z_2 = Z_c \parallel Z_L = \frac{-j 25 \times j 20}{-j 5} = j 100 \Omega.$$

$$V_o = 20 \angle -15^\circ \times \frac{Z_2}{Z_1 + Z_2}$$

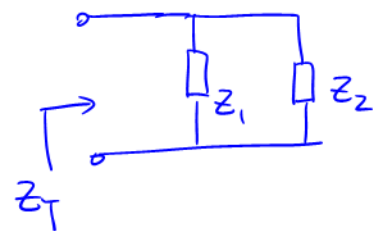
$$= 20 \angle -15^\circ \times \frac{j 100}{60 + j 100} = 16.49 + j 4.72 \text{ V.}$$

Example 8: impedance, KCL and KVL

Find Z_T and V_{ab} .



$$\begin{aligned}
 Z_T &= Z_1 \parallel Z_2 \\
 &= \frac{(20 - j5)(40 + j10)}{60 + j5} \\
 &= \frac{408}{29} - j\frac{34}{29} = \underline{14.07 - j1.17 \Omega.} \\
 &\quad R > 0.
 \end{aligned}$$



$$\begin{aligned}
 V_{ab} &= V_a - V_b, \quad V_a = V_s \cdot \frac{-j5}{20 - j5}, \quad V_b = V_s \cdot \frac{40}{40 + j10} \\
 &= V_s \cdot \left(\frac{-j5}{20 - j5} - \frac{40}{40 + j10} \right) \\
 &= j115 \cdot \left(-\frac{15}{17} \right) = \underline{-j101.47 \text{ V.}}
 \end{aligned}$$